

PAPER_1759 — Cosmological Baryon Fraction = $f_{\text{bar}} = \Phi_{5/6} \times \beta_i = 0.502$ (matches Planck $f_b \approx 0.50$)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Cosmology **Date:** June 18, 2026 **Status:** CLOSED — 0.7% closure (PAPER_132x-135x astrophysics + condensed matter)

Observation

PAPER_1329 (PAPER_132x-135x corpus) gives:

$$f_{\text{bar}} = \Phi_{5/6} \times \beta_i = 0.502 \text{ (matches Planck } f_b \approx 0.50)$$

Status: **0.7%**.

UQFF Closed Identity

$$f_{\text{bar}} = \Phi_{5/6} \times \beta_i = 0.502 \text{ (matches Planck } f_b \approx 0.50) \quad 0.7\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1329
- Related: PAPER_1746-1755 (tier-33); PAPER_1736-1745 (tier-32)
- Calculator dispatch: `calculate_paradox({"paradox": "baryon_fraction_50_6"})`

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